Real-time and efficient multi-target speech extraction via target-direction-agnostic feature extraction and late steering-vector fusion

Cheng Chi,[1,2] Yuxuan Ke,[1,2] Andong Li,[1,2] Dan Rao,[3] Xiaodong Li,[1,2] and Chengshi Zheng[1,2,a)]

[1)] *Key Laboratory of Noise and Vibration Research, Institute of Acoustics, Chinese Academy of Sciences, Beijing 100190, China*

[2)] *University of Chinese Academy of Sciences, Beijing 100049, China*

[3)] *School of Physics and Optoelectronics, South China University of Technology, Guangzhou, Guangdong 510641, China*

chicheng2021@mail.ioa.ac.cn, keyuxuan@mail.ioa.ac.cn, liandong@mail.ioa.ac.cn, phdrao@scut.edu.cn, lxd@mail.ioa.ac.cn, cszheng@mail.ioa.ac.cn

Abstract: Existing neural beamformers guided by a target direction of arrival (DOA) typically introduce this directional information during feature extraction, requiring repeated inference for multiple targets. Inspired by minimum power distortionless response beamforming, the proposed framework first extracts and embeds discriminative features without DOA priors and subsequently incorporates this directional information during beamforming-weight estimation. Specifically, a lightweight module estimates target-dependent weights from the shared features. Experiments show that the framework reduces computation for multi-target speech extraction while maintaining comparable performance and offering flexibility in the number of targets.

[Editor: —] https://doi.org/

Date: 28 October 2026

1. Introduction

Multichannel multi-target speech extraction recovers the speech of multiple requested speakers from microphone-array signals containing competing speech, noise, and reverberation.[1,2] A target cue specifies the source to be extracted and should be available before implementing the extraction to enable real-time processing.[3] A target direction of arrival (DOA) provides such a cue[1] and can be obtained from an audiovisual localization system[4] or a predefined position.[5]

To extract target speech based on directional cues, conventional adaptive beamformers compute spatial filtering weights and apply them to the multichannel mixture, including minimum variance distortionless response (MVDR),[6] minimum power distortionless response (MPDR),[7] and linearly constrained minimum variance (LCMV)[8] beamformers. For multiple targets, MVDR requires an interference-plus-noise power spectral density (PSD) matrix for each target, obtained by subtracting the target PSD matrix from the noisy PSD matrix. In contrast, MPDR and LCMV use the noisy PSD matrix directly, combining it with the corresponding steering vector or linear constraints to obtain target-specific weights.

Beyond conventional beamforming, neural DOA-informed methods commonly receive the multichannel mixture and target direction as network inputs. The temporal-spatial neural filter[9] combines temporal, spectral, and spatial features. In the joint spatial and tempo-spectral nonlinear filter (JNF), configured as a spatially selective filter (JNF-SSF),[1] the target DOA initializes the first recurrent layer. Location-dependent feature extraction[10] conditions the first convolutional encoder layer on the requested location, while deep informed spatio-spectral filtering (DISF)[11] fuses steering information in its first stage. Position-aware target speaker extraction (PATSE)[12] forms reusable target-direction-agnostic audio features, yet still evaluates its spatial encoder, conditioner, separator, and decoder per target DOA. Consequently, these methods make part or all subsequent processing target dependent. A multiple-input multiple-output (MIMO) recurrent beamformer[13] instead processes all target directions in one evaluation, but fixes the output count in the network architecture. Other work reduces single-network complexity via deep filtering,[14] spiking or hybrid architectures,[15,16] band-split recurrence,[17] and cross-modal distillation,[18] but is not specifically optimized for multi-speaker extraction.

To handle a variable number of targets without repeated estimation of target-dependent features, we propose an MPDR-inspired architecture that separates target-direction-agnostic feature extraction from target-dependent weight generation. In this proposed architecture, a shared neural front end, which serves as a feature encoder, computes the array-signal representation only once, and a lightweight neural head then combines it with each steering vector to generate complex-valued beamforming weights. We implement the design with a causal lightweight backbone and adapt the

[a)] Author to whom correspondence should be addressed.

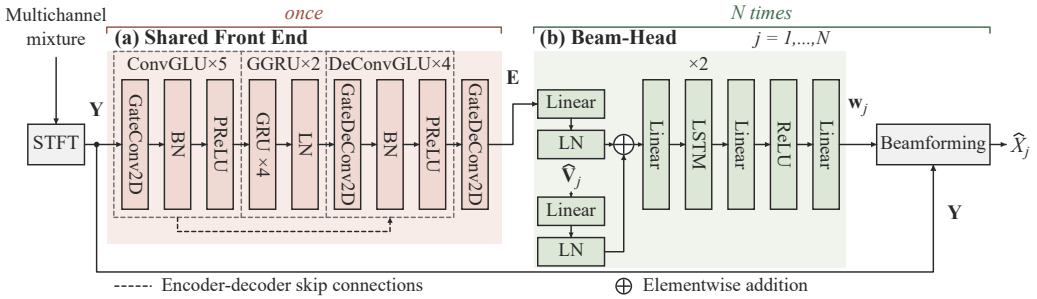


Fig. 1. Overall architecture of the proposed multi-target speech extraction framework.

embedding-and-beamforming network (EaBNet),[19] TaylorBeamformer,[20] and JNF-SSF[1,21] to the same structure. Experiments with two or three speakers assess extraction quality as well as how runtime and operation counts scale with the number of targets.

2. Target-direction-agnostic feature sharing and late steering-vector fusion

2.1 MPDR-inspired computational decomposition

Consider an array of M microphones observing P speech sources in noisy and reverberant environments, where M and P denote the numbers of microphones and speech sources, respectively. At frequency bin k and frame l, the microphone signals $\mathbf{y}(k,l) \in \mathbb{C}^{M\times 1}$ satisfy

$$\mathbf{y}(k,l) = \sum_{p=1}^{P} \mathbf{a}_p(k) S_p(k,l) + \mathbf{n}(k,l), \tag{1}$$

where $S_p(k,l)$ is the short-time Fourier transform (STFT) coefficient of the pth speech source, $\mathbf{a}_p(k) \in \mathbb{C}^{M\times 1}$ is its acoustic transfer vector, and $\mathbf{n}(k,l) \in \mathbb{C}^{M\times 1}$ is the additive noise. With a complex beamforming weight vector $\mathbf{w}_j(k,l) \in \mathbb{C}^{M\times 1}$ for the jth target speaker, where $j \in \{1,\ldots,N\}$ indexes the $N \le P$ requested targets, we can obtain the target estimate $\widehat{X}_j(k,l) = \mathbf{w}_j^H(k,l)\mathbf{y}(k,l)$, where $(\cdot)^H$ denotes the conjugate transpose.

Given the noisy PSD matrix $\mathbf{R}_{yy}(k,l) = \mathbb{E}\{\mathbf{y}(k,l)\mathbf{y}^H(k,l)\}$ and the target steering vector $\mathbf{v}_j(k) \in \mathbb{C}^{M\times 1}$ determined by the target DOA and array geometry, the MPDR beamformer is

$$\mathbf{w}_{j,\mathrm{MPDR}}(k,l) = \frac{\mathbf{R}_{yy}^{-1}(k,l)\mathbf{v}_j(k)}{\mathbf{v}_j^H(k)\mathbf{R}_{yy}^{-1}(k,l)\mathbf{v}_j(k)}. \tag{2}$$

Importantly, the acoustic transfer vector $\mathbf{a}_j(k)$ in Eq. (1) includes the early-and-late reflections of the room, which is quite different from $\mathbf{v}_j(k)$ in reverberant environments. The inverse matrix $\mathbf{R}_{yy}^{-1}(k,l)$ in Eq. (2) is determined by the noisy mixture alone and is therefore common to all target speakers, whereas target dependence is confined to the steering-vector terms. The proposed neural beamformer adopts this computational decomposition, yet avoids explicitly estimating or inverting $\mathbf{R}_{yy}$.

2.2 Late steering-vector fusion and target-dependent beamforming

Inspired by the traditional MPDR beamformer, we first analyze and extract acoustic scene features through a target-direction-agnostic feature extractor. As illustrated in Fig. 1, the shared front-end module B_ψ in Fig. 1(a) serves as this feature extractor and produces an array-signal representation across targets, while a parameter-shared beamforming head f_ϕ in Fig. 1(b) combines this representation with each steering vector to generate target-specific weights, using linear projections to adapt backbone-specific feature dimensions. Let $\mathbf{Y} \in \mathbb{C}^{F\times T\times M}$ collect the M-channel STFT coefficients over F frequency bins and T frames. The two-stage neural beamformer is decomposed as

$$\mathbf{E} = B_\psi(\mathbf{Y}), \qquad \mathbf{w}_j = f_\phi\left(\mathbf{E}, \widehat{\mathbf{V}}_j\right). \tag{3}$$

In Eq. (3), B_ψ runs once per mixture to encode $\mathbf{Y}$ into a D_E-dimensional representation $\mathbf{E} \in \mathbb{R}^{F\times T\times D_E}$. The input DOA and array geometry define a far-field plane-wave steering vector $\widehat{\mathbf{v}}_j(k) \in \mathbb{C}^{M\times 1}$ at each frequency, and stacking them gives $\widehat{\mathbf{V}}_j = [\widehat{\mathbf{v}}_j(1),\ldots,\widehat{\mathbf{v}}_j(F)]^\mathsf{T} \in \mathbb{C}^{F\times M}$. Note that $\widehat{\mathbf{v}}_j(k)$ is an estimate of $\mathbf{v}_j(k)$, and this estimate is determined by the target direction estimation. Only the steering input $\widehat{\mathbf{V}}_j$ of the shared head f_ϕ changes with j when estimating $\mathbf{w}_j$, which collects $\mathbf{w}_j(k,l)$ over all frequency bins and frames.

Within the beamforming head in Fig. 1(b), the shared representation and steering vector are projected to a common D_H-dimensional space before fusion. Layer normalization (LN) is applied separately to the two projected representations,

and they are then combined through elementwise addition, given by

$$\mathbf{F}_j(k,l) = \mathrm{LN}(\mathbf{W}_E\mathbf{E}(k,l) + \mathbf{b}_E) + \alpha\,\mathrm{LN}\left(\mathbf{W}_v \begin{bmatrix} \Re\{\widehat{\mathbf{v}}_j(k)\} \\ \Im\{\widehat{\mathbf{v}}_j(k)\} \end{bmatrix} + \mathbf{b}_v\right), \tag{4}$$

where $\mathbf{W}_E \in \mathbb{R}^{D_H \times D_E}$ and $\mathbf{W}_v \in \mathbb{R}^{D_H \times 2M}$ are backbone-specific projections with biases $\mathbf{b}_E \in \mathbb{R}^{D_H \times 1}$ and $\mathbf{b}_v \in \mathbb{R}^{D_H \times 1}$, and $\alpha \in \{0, 1\}$ equals 1 except when evaluating the model without steering input ($\alpha = 0$). A linear adapter, stacked unidirectional long short-term memory (LSTM) layers, and two linear output layers separated by a rectified linear unit (ReLU) then estimate the $2M$ real and imaginary components of $\mathbf{w}_j(k,l)$, which filter $\mathbf{Y}$ to obtain $\widehat{X}_j$.

2.3 Lightweight shared front end

To achieve low complexity for real-time operation, the shared front end B_ψ in Fig. 1(a) is designed as a causal U-shaped encoder–decoder based on convolutional gated linear unit (ConvGLU), grouped gated recurrent unit (GGRU), and deconvolutional gated linear unit (DeConvGLU) blocks. The encoder stacks ConvGLU blocks to downsample the frequency dimension of the real and imaginary input STFT components, where each block comprises a gated two-dimensional convolution (GateConv2D), batch normalization (BN), and a parametric rectified linear unit (PReLU).

At the bottleneck, stacked GGRU stages split the feature sequence across parallel single-layer gated recurrent units (GRUs) and concatenate their outputs before LN.[14] Skip connections link corresponding encoder and decoder levels. The decoder stacks DeConvGLU blocks, each comprising a gated two-dimensional transposed convolution (GateDeConv2D), BN, and PReLU, followed by a final GateDeConv2D block without normalization or activation that outputs $\mathbf{E}$. Unit temporal kernels and unidirectional recurrence ensure frame-level causality.

2.4 Training objective

To train the shared front end and beamforming head in an end-to-end manner, the proposed network is optimized in the compressed complex spectral domain. With $\mathcal{C}_\gamma(X) = |X|^\gamma \exp(\mathrm{i}\angle X)$, let X_j and $\widehat{X}_j$ denote the STFT coefficients of the early-reflected target speech, namely the early component of target j at the reference microphone, and its estimate. The loss for target j is

$$\mathcal{L}_j = \left\|\mathcal{C}_\gamma(\widehat{X}_j) - \mathcal{C}_\gamma(X_j)\right\|_F^2 + \left\|\left|\mathcal{C}_\gamma(\widehat{X}_j)\right| - |\mathcal{C}_\gamma(X_j)|\right\|_F^2, \qquad \gamma = 0.5, \tag{5}$$

where the unit-weight terms in Eq. (5) measure complex-spectrum and compressed-magnitude errors using the Frobenius norm $\|\cdot\|_F$.[22,23] The target-averaged objective is $(1/N)\sum_{j=1}^{N}\mathcal{L}_j$.

3. Experimental setup

3.1 Data and acoustic simulation

To evaluate the proposed framework under reverberant multi-speaker conditions, we simulate mixtures by convolving clean speech and noise from the Deep Noise Suppression (DNS) Challenge corpus[24] with room impulse responses generated via the image-source method.[25] A six-microphone circular array of 5.0-cm radius is placed in rooms with lengths and widths of 5–10 m and heights of 3–4 m. The reverberation time T_{60} spans 0.1–0.7 s, and source distances from the array center span 1–3 m. Source azimuths are sampled on a 5° grid between -90° and 90° with at least 30° separation between concurrent speakers, whereas noise directions are unconstrained. The early-reflected target speech comprises the direct path and reflections arriving within 20 ms after the direct path.

Building on this setup, each one- or two-speaker training set contains 6600 independently simulated 30-s mixtures, and each validation set contains 740 mixtures. The single-speaker ablation trains and evaluates on one speaker, whereas multi-target systems train on two speakers and evaluate on two or three speakers, extracting all active sources so that $N = P$. Each speaker-count and signal-to-noise ratio (SNR) condition contains 370 test mixtures of 30 s. Within each speaker count, the four SNRs share speech and room configurations and differ only in added-noise level. The nominal SNRs of 0, 5, 10, and 15 dB are defined as the power ratio of the summed fully reverberant speech of all active speakers to the non-speech additive noise, averaged over time and all six microphones. At the reference microphone, the reverberant signal-to-interference ratio ranges from about -4 to $+4$ dB for two speakers and from -6.3 to $+1.1$ dB for three speakers, with a mean of -3.0 dB. Noise is added in successive non-overlapping 10-s segments. Each segment contains one point source, convolved with the room impulse response of a position drawn uniformly inside the room and at least 0.5 m from the walls.

3.2 Systems and training

Because the shared front end is a flexible and replaceable module, we also incorporate the backbones of EaBNet,[19] causal zeroth-order TaylorBeamformer,[20] and JNF-SSF[1] into the proposed framework, while evaluating original DISF[11] and JNF-SSF[1] separately per requested speaker as target-DOA-conditioned baselines. Lightweight, EaBNet, and TaylorBeamformer operate causally without future spectral frames, whereas JNF-SSF evaluates generalization to a noncausal backbone.

Each shared backbone runs once per mixture. In the Lightweight system, the shared front end uses five ConvGLU blocks, two GGRU stages that each split a 256-dimensional sequence across four 64-unit GRUs, four DeConvGLU blocks, and a final GateDeConv2D block, all with 64 channels, $(1, 2)$ frequency strides, and $(1, 3)$ kernels except $(1, 5)$ in the first

and last blocks. Its head sets $D_E = D_H = 64$ with identity audio and adapter mappings, a 12-to-64 steering projection, two 64-unit LSTM layers, and 64- and $2M$-unit output layers. EaBNet shares its U^2 encoder, squeezed temporal convolutional network (TCN), and decoder, and TaylorBeamformer shares the corresponding path. Their $F \times T \times 64$ features and steering inputs are projected to 64 dimensions before the same head structure. JNF-SSF shares its frequency- and time-axis LSTMs, 128-dimensional projection, and normalization. Its $F \times T \times 128$ representation and steering vector are projected to 128 dimensions before an adapter, a bidirectional beam LSTM, and 96- and $2M$-unit output layers. This variant retains the original JNF-SSF signal representation and optimization.[1,21]

All signals are sampled at 16 kHz, and Lightweight, EaBNet, and TaylorBeamformer use a 320-sample (20-ms) Hann window with 50% overlap and a 10-ms hop, together with a 320-point STFT yielding 161 bins. They use no future frames, so the STFT and overlap-add delay equals this 20-ms window length. Their inputs are the real and imaginary components of $\mathcal{C}_{0.5}(\mathbf{Y})$, while spatial filtering uses the uncompressed STFT coefficients. Lightweight, EaBNet, and TaylorBeamformer train end to end from random initialization on identical data within each experiment, using 4.0-s segments for 100 epochs, Adam with initial learning rate 5×10^{-4} and weight decay 10^{-7}, gradient-norm clipping at 5.0, and the lowest-validation-loss checkpoint. Target azimuths are perturbed by $\mathcal{U}(-5^\circ, 5^\circ)$ during training, whereas standard evaluation uses true azimuths.

3.3 Evaluation metrics and computational cost

To quantify extraction quality, we use wideband perceptual evaluation of speech quality (PESQ),[26,27] extended short-time objective intelligibility (ESTOI),[28] scale-invariant signal-to-distortion ratio (SI-SDR),[29] and the Deep Noise Suppression mean opinion score (DNSMOS).[30] PESQ, ESTOI, and SI-SDR use the early-reflected target speech as the reference, whereas DNSMOS estimates overall quality (OVRL), speech quality (SIG), and background noise quality (BAK) without a reference. The noisy-mixture baseline evaluates the unprocessed reference-microphone signal for every target. Scores are averaged over targets and mixtures, and across the 370 test configurations averaged over targets and SNRs, standard deviations (SDs) of processed systems are about 3–4 dB for SI-SDR, 0.3–0.4 for PESQ, and 0.1 for ESTOI.

Table 1: Reference-based and DNSMOS (OVRL, SIG, BAK) scores averaged over targets and the 0, 5, 10, and 15 dB SNR conditions. Boldface marks the highest mean per metric and speaker count within each comparison.

Method / setting	Speakers	PESQ	ESTOI	SI-SDR (dB)	OVRL	SIG	BAK
Noisy mixture	1	2.27	0.609	1.52	1.81	2.19	2.46
Without steering vector ($\alpha = 0$)	1	2.90	0.764	5.74	2.39	2.77	4.08
Early steering-vector fusion	1	**3.02**	**0.784**	**6.45**	**2.45**	**2.83**	**4.09**
Late steering-vector fusion ($\alpha = 1$)	1	2.95	0.771	6.04	2.41	2.79	4.08
Early steering-vector fusion (matched)	2	2.62	0.671	3.82	**2.27**	**2.65**	**4.05**
Late steering-vector fusion (matched)	2	**2.64**	**0.673**	**4.20**	2.25	2.62	4.04
Noisy mixture	2	1.80	0.380	−4.28	1.44	1.70	1.75
DISF	2	2.57	0.667	4.20	**2.44**	**2.84**	**4.09**
JNF-SSF	2	2.64	0.682	5.08	2.25	2.63	4.00
JNF-SSF (shared front end)	2	**2.72**	**0.717**	**6.34**	2.35	2.73	4.03
EaBNet (shared front end)	2	2.62	0.671	4.34	2.29	2.66	4.05
TaylorBeamformer (shared front end)	2	2.65	0.674	4.16	2.28	2.65	4.05
Lightweight (shared front end)	2	2.63	0.668	4.01	2.26	2.63	4.04
Noisy mixture	3	1.66	0.277	−6.56	1.30	1.51	1.50
DISF	3	2.32	0.588	2.40	**2.39**	**2.79**	**4.08**
JNF-SSF	3	2.38	0.593	3.26	2.10	2.48	3.95
JNF-SSF (shared front end)	3	**2.39**	**0.620**	**4.19**	2.21	2.59	3.97
EaBNet (shared front end)	3	2.28	0.572	2.22	2.13	2.49	3.99
TaylorBeamformer (shared front end)	3	2.36	0.585	2.29	2.13	2.49	4.00
Lightweight (shared front end)	3	2.37	0.588	2.35	2.10	2.45	3.99

In addition to extraction quality, operation counts are reported in giga multiply-accumulate operations per second (GMAC/s). Offline real-time factor (RTF) is measured on a single core of an Intel Core i9-10900 CPU as the median over 40 runs of a 1-s six-channel mixture after 10 warm-ups, covering STFT, neural inference, spatial filtering, and inverse STFT. For the causal systems, we also report the 95th-percentile (P95) streaming output delay on the same CPU using 10- and 40-ms input packets.

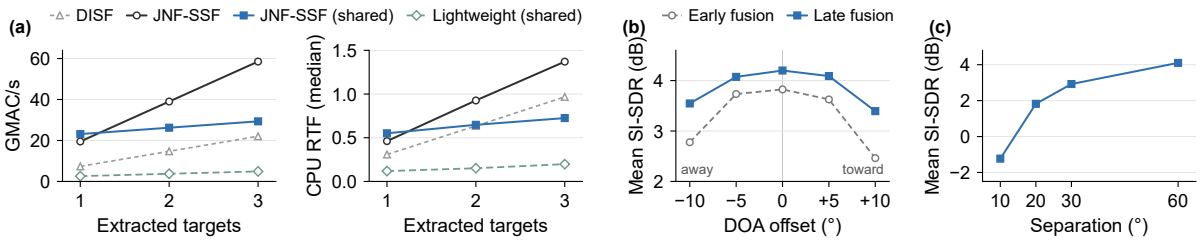


Fig. 2. Computational scaling and two-speaker robustness. (a) GMAC/s (left) and CPU RTF (right) versus extracted target count. (b) Mean SI-SDR of the matched two-source early- and late-fusion Lightweight models under DOA offsets away from (−) or toward (+) the interferer. (c) Mean SI-SDR of the late-fusion model without DOA offset versus speaker separation, with both speakers extracted in (b) and (c).

4. Results and discussion

4.1 Extraction quality

To evaluate the effect of the steering-vector fusion point in the Lightweight architecture, we first conduct a one-speaker ablation. Early fusion injects the normalized steering representation after the first ConvGLU block. Late fusion ($\alpha = 1$) follows Eq. (4). The model without steering input ($\alpha = 0$) is trained separately from the same initial weights. In the first four rows of Table 1, early fusion achieves the highest single-speaker scores. Late fusion keeps the front end direction agnostic and still outperforms the $\alpha = 0$ baseline on all three reference-based metrics, raising SI-SDR from 5.74 to 6.04 dB. The steering vector therefore remains informative when it is introduced only at the beamforming head.

To further test whether delaying steering fusion impairs target selection under competing speech, we compare matched two-source early- and late-fusion Lightweight models. They share the initial weights, training data, head structure, and optimization settings, and differ only in where the steering representation enters the network. In the fifth and sixth rows of Table 1, late fusion obtains a higher SI-SDR than early fusion (4.20 vs 3.82 dB), while PESQ and ESTOI remain nearly identical. The single-speaker advantage of early fusion therefore does not carry over to competing speech. A shared front end can preserve the spatial cues of all active sources.

To assess generalizability across backbones and speaker counts, we compare the shared-front-end variants of Lightweight, EaBNet, TaylorBeamformer, and JNF-SSF with the per-target baselines DISF and JNF-SSF on two- and three-speaker mixtures. In the remaining rows of Table 1, shared-front-end JNF-SSF achieves the highest mean PESQ, ESTOI, and SI-SDR. Its SI-SDR exceeds that of repeated JNF-SSF by 1.26 dB for two speakers and 0.93 dB for three speakers across all SNR conditions. The other three shared systems obtain SI-SDR scores within 0.2 dB of DISF for both speaker counts. DISF achieves the highest DNSMOS scores because DNSMOS lacks a target reference and cannot distinguish target speech from interfering speech. Adding another interfering speaker reduces the scores of all systems, because each target then faces two interferers.

4.2 Inference cost versus target count

To evaluate how computational cost scales with the number of targets N, we compare parameter counts and operation counts (GMAC/s) for $N = 1, 2, 3$ in Fig. 2(a) and Table 2. The plain Lightweight row is a separate audio-only baseline without the steering branch. Head parameters are stored once, so the parameter count does not grow with the number of targets. Relative to the original systems, TaylorBeamformer adds an 832-parameter steering projection, Lightweight widens the LSTM input and drops the audio projections, EaBNet expands its TCN from 18 to 32 blocks, and JNF-SSF replaces conditioning and mask output with a shared feature projection and a beamforming head. Rerunning DISF and JNF-SSF adds about 7.4 and 19.5 GMAC/s per target. The shared systems evaluate the front end once and add only the head, which requires about 1.16 GMAC/s per target for Lightweight, EaBNet, and TaylorBeamformer and 3.10 GMAC/s per target for JNF-SSF. The front end accounts for 55% (Lightweight) to 87% (JNF-SSF) of the single-target shared cost, so sharing it reduces the operation count from two targets onward. Three-target JNF-SSF saves 49.9%. Three-target late fusion saves 36% relative to early fusion (4.90 vs 7.62 GMAC/s).

To further verify runtime efficiency and real-time streaming capability, we measure the single-core offline RTF and the streaming P95 output delay for 10- and 40-ms input packets at $N = 1, 2, 3$. Fig. 2(a) and Table 2 show the same crossover in CPU runtime as in GMAC/s. Three-target shared JNF-SSF reduces the offline RTF of repeated JNF-SSF from 1.370 to 0.725, a 47.1% reduction. The causal matched early- and late-fusion models reach three-target offline RTFs of 0.348 and 0.192. With three targets, the late-fusion model produces its first output at 26.45 ms and has a P95 delay of 24.27 ms. Both values include the 20-ms window length, and the P95 per-packet processing time is 4.20 ms. All tested causal systems run in real time for both packet sizes. DISF and both JNF-SSF variants are noncausal and are excluded from the streaming test.

Table 2. Parameters, GMAC/s, single-thread offline RTF, and streaming P95 delay (ms, including the 20-ms window length) for 10- and 40-ms input packets ($N = 1, 2, 3$ targets). Dashes (–) denote settings without target-direction input for $N > 1$ or noncausal systems excluded from streaming.

System	Parameters (M)	GMAC/s			Offline RTF			10-ms packets			40-ms packets		
		1	2	3	1	2	3	1	2	3	1	2	3
DISF	5.77	7.35	14.71	22.06	0.308	0.635	0.967	–	–	–	–	–	–
JNF-SSF	1.23	19.51	39.01	58.52	0.461	0.926	1.370	–	–	–	–	–	–
EaBNet	2.83	4.62	–	–	0.170	–	–	27.35	–	–	62.35	–	–
TaylorBeamformer	2.37	4.32	–	–	0.171	–	–	29.69	–	–	64.52	–	–
Lightweight	0.79	2.46	–	–	0.118	–	–	23.54	–	–	56.67	–	–
JNF-SSF (shared)	1.44	23.13	26.23	29.34	0.551	0.648	0.725	–	–	–	–	–	–
EaBNet (shared)	3.87	4.74	5.89	7.05	0.176	0.224	0.260	28.43	28.84	29.23	64.07	65.66	67.24
TaylorBeamformer (shared)	2.37	4.34	5.49	6.65	0.163	0.207	0.251	29.55	29.75	29.90	64.38	65.81	67.21
Lightweight (shared)	0.79	2.59	3.75	4.90	0.118	0.151	0.198	23.52	23.92	24.29	56.72	58.22	59.68
Early fusion (matched)	0.79	2.58	5.10	7.62	0.122	0.238	0.348	23.51	25.05	25.89	56.80	61.11	64.39
Late fusion (matched)	0.79	2.59	3.75	4.90	0.116	0.157	0.192	23.53	23.92	24.27	56.72	58.28	59.72

4.3 Robustness to DOA mismatch

To investigate robustness against DOA estimation errors, we first test the single-speaker late-fusion Lightweight model without retraining under fixed offsets of 0°, $\pm5^\circ$, and $\pm10^\circ$. SI-SDR drops by at most 0.06 dB within $\pm5^\circ$ and by 0.23 dB at $\pm10^\circ$. We then evaluate the matched two-source early- and late-fusion models under 5° and 10° offsets away from and toward the interfering speaker, scoring each output against the requested target without permutation. Fig. 2(b) shows that SI-SDR decreases for both models as the offset increases, especially at 10° toward the interferer. Late fusion remains above early fusion at every offset, and its largest loss at $\pm10^\circ$ is smaller (0.81 vs 1.36 dB). Delaying steering fusion to the head therefore preserves target selection under moderate DOA errors.

Finally, to evaluate the spatial resolution limit for closely spaced speakers, we test the late-fusion model without a DOA offset on 100 held-out two-speaker configurations matched across separations of 10°, 20°, 30°, and 60°. Fig. 2(c) shows that the mean SI-SDR decreases steadily as the speakers approach each other. The score becomes negative at a 10° separation, where the two steering vectors are nearly parallel.

5. Conclusion

The proposed MPDR-inspired decomposition computes target-direction-agnostic array features only once for all targets through a shared front end and combines them with each steering vector in a lightweight per-target head. Each additional target therefore requires only one head evaluation rather than a complete network evaluation, so the computational saving increases with the number of targets. Across four backbones, the shared-front-end systems maintain or improve extraction quality relative to target-conditioned baselines, where the noncausal shared JNF-SSF achieves the best reference-based scores and the causal shared Lightweight model operates in real time with an STFT and overlap-add delay equal to the 20-ms window length. Furthermore, the single-speaker Lightweight model incurs at most a 0.23-dB SI-SDR loss under DOA offsets up to $\pm10^\circ$, and in the matched two-source comparison, late fusion outperforms early fusion across all tested DOA errors, although closely spaced speakers remain a limitation. Overall, this decomposition extends both DOA-informed and DOA-agnostic single-target backbones to efficient multi-target speech extraction.

Author Declarations

Conflict of Interest. The authors have no conflicts to disclose.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References and links

[1] K. Tesch and T. Gerkmann, "Multi-channel speech separation using spatially selective deep non-linear filters," IEEE/ACM Trans. Audio Speech Lang. Process. **32**, 542–553 (2024) doi: 10.1109/TASLP.2023.3334101.

[2] T. Serre, M. Fontaine, E. Benhaim, and S. Essid, "MTSE: Multi-target speaker extraction for conversation scenarios," in *Proceedings of Interspeech 2025*, Rotterdam, The Netherlands (August 17–21, 2025), pp. 2970–2974, doi: 10.21437/Interspeech.2025-2077.

[3] H. Sato, T. Moriya, M. Mimura, *et al.*, "SpeakerBeam-SS: Real-time target speaker extraction with lightweight Conv-TasNet and state space modeling," in *Proceedings of Interspeech 2024*, Kos, Greece (September 1–5, 2024), pp. 5033–5037, doi: 10.21437/Interspeech.2024-413.

[4] X. Qian, Q. Zhang, G. Guan, and W. Xue, "Deep audio-visual beamforming for speaker localization," IEEE Signal Process. Lett. **29**, 1132–1136 (2022) doi: 10.1109/LSP.2022.3165466.

[5] H. Taherian, K. Tan, and D. Wang, "Multi-channel talker-independent speaker separation through location-based training," IEEE/ACM Trans. Audio Speech Lang. Process. **30**, 2791–2800 (2022) doi: 10.1109/TASLP.2022.3202129.

[6] M. Souden, J. Benesty, and S. Affes, "A study of the LCMV and MVDR noise reduction filters," IEEE Trans. Signal Process. **58**(9), 4925–4935 (2010) doi: 10.1109/TSP.2010.2051803.

[7] S. Gannot, E. Vincent, S. Markovich-Golan, and A. Ozerov, "A consolidated perspective on multimicrophone speech enhancement and source separation," IEEE/ACM Trans. Audio Speech Lang. Process. **25**(4), 692–730 (2017) doi: 10.1109/TASLP.2016.2647702.

[8] O. L. Frost, III, "An algorithm for linearly constrained adaptive array processing," Proc. IEEE **60**(8), 926–935 (1972) doi: 10.1109/PROC.1972.8817.

[9] R. Gu and Y. Zou, "Temporal-spatial neural filter: Direction informed end-to-end multi-channel target speech separation," arXiv preprint arXiv:2001.00391 (2020) doi: 10.48550/arXiv.2001.00391.

[10] A. Bohlender, A. Spriet, W. Tirry, and N. Madhu, "Spatially selective speaker separation using a DNN with a location dependent feature extraction," IEEE/ACM Trans. Audio Speech Lang. Process. **32**, 930–945 (2024) doi: 10.1109/TASLP.2023.3343605.

[11] X. Luo, Y. Ke, X. Li, and C. Zheng, "Deep informed spatio-spectral filtering for multi-channel speech extraction against steering vector uncertainties," Appl. Acoust. **228**, 110259 (2025) doi: 10.1016/j.apacoust.2024.110259.

[12] Y. Wang, J. Chen, W. Zhou, and T. Kawahara, "Position-aware target speaker extraction for long-form multi-party conversations: A diarization-free framework for ASR," arXiv preprint arXiv:2606.29497 (2026) doi: 10.48550/arXiv.2606.29497.

[13] X. Li, Y. Xu, M. Yu, *et al.*, "MIMO self-attentive RNN beamformer for multi-speaker speech separation," in *Proceedings of Interspeech 2021*, Brno, Czech Republic (August 30–September 3, 2021), pp. 1119–1123, doi: 10.21437/Interspeech.2021-570.

[14] H. Schröter, A. N. Escalante-B., T. Rosenkranz, and A. Maier, "DeepFilterNet: A low complexity speech enhancement framework for full-band audio based on deep filtering," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Singapore (May 23–27, 2022), pp. 7407–7411, doi: 10.1109/ICASSP43922.2022.9747055.

[15] E. Liu, A. Li, C. Fan, *et al.*, "SSE-Net: Toward low-power-consumption spiking neural network for monaural speech enhancement," IEEE Trans. Audio Speech Lang. Process. **34**, 2077–2089 (2026) doi: 10.1109/TASLPRO.2026.3677621.

[16] C. Fan, E. Liu, J. Zhou, *et al.*, "DBHN-Net: Dual-branch hybrid neural network for low-complexity monaural speech enhancement," IEEE Trans. Pattern Anal. Mach. Intell. **48**, 12196–12210 (2026).

[17] J. Yu, H. Chen, Y. Luo, *et al.*, "High fidelity speech enhancement with band-split RNN," in *Proceedings of Interspeech 2023*, Dublin, Ireland (August 20–24, 2023), pp. 2483–2487, doi: 10.21437/Interspeech.2023-1433.

[18] C. Fan, W. Xiang, J. Tao, *et al.*, "Cross-modal knowledge distillation with multi-stage adaptive feature fusion for speech separation," IEEE Trans. Audio Speech Lang. Process. **33**, 935–948 (2025) doi: 10.1109/TASLPRO.2025.3533359.

[19] A. Li, W. Liu, C. Zheng, and X. Li, "Embedding and beamforming: All-neural causal beamformer for multichannel speech enhancement," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Singapore (May 23–27, 2022), pp. 6487–6491, doi: 10.1109/ICASSP43922.2022.9746432.

[20] A. Li, G. Yu, C. Zheng, and X. Li, "TaylorBeamformer: Learning all-neural beamformer for multi-channel speech enhancement from Taylor's approximation theory," in *Proceedings of Interspeech 2022*, Incheon, Korea (September 18–22, 2022), pp. 5413–5417, doi: 10.21437/Interspeech.2022-159.

[21] K. Tesch and T. Gerkmann, "Insights into deep non-linear filters for improved multi-channel speech enhancement," IEEE/ACM Trans. Audio Speech Lang. Process. **31**, 563–575 (2023) doi: 10.1109/TASLP.2022.3221046.

[22] A. Li, W. Liu, C. Zheng, *et al.*, "Two heads are better than one: A two-stage complex spectral mapping approach for monaural speech enhancement," IEEE/ACM Trans. Audio Speech Lang. Process. **29**, 1829–1843 (2021) doi: 10.1109/TASLP.2021.3079813.

[23] A. Li, C. Zheng, R. Peng, and X. Li, "On the importance of power compression and phase estimation in monaural speech dereverberation," JASA Express Lett. **1**(1), 014802 (2021) doi: 10.1121/10.0003321.

[24] C. K. A. Reddy, H. Dubey, K. Koishida, *et al.*, "Interspeech 2021 Deep Noise Suppression challenge," in *Proceedings of Interspeech 2021*, Brno, Czech Republic (August 30–September 3, 2021), pp. 2796–2800, doi: 10.21437/Interspeech.2021-1609.

[25] J. B. Allen and D. A. Berkley, "Image method for efficiently simulating small-room acoustics," J. Acoust. Soc. Am. **65**(4), 943–950 (1979) doi: 10.1121/1.382599.

[26] A. W. Rix, J. G. Beerends, M. P. Hollier, and A. P. Hekstra, "Perceptual evaluation of speech quality (PESQ)—a new method for speech quality assessment of telephone networks and codecs," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Salt Lake City, UT (May 7–11, 2001), Vol. 2, pp. 749–752, doi: 10.1109/ICASSP.2001.941023.

[27] ITU-T, "Wideband extension to Recommendation P.862 for the assessment of wideband telephone networks and speech codecs," Recommendation P.862.2, International Telecommunication Union (2007), https://www.itu.int/rec/T-REC-P.862.2-200711-W/en.

[28] J. Jensen and C. H. Taal, "An algorithm for predicting the intelligibility of speech masked by modulated noise maskers," IEEE/ACM Trans. Audio Speech Lang. Process. **24**(11), 2009–2022 (2016) doi: 10.1109/TASLP.2016.2585878.

[29] J. Le Roux, S. Wisdom, H. Erdogan, and J. R. Hershey, "SDR–half-baked or well done?," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Brighton, UK (May 12–17, 2019), pp. 626–630, doi: 10.1109/ICASSP.2019.8683855.

[30] C. K. A. Reddy, V. Gopal, and R. Cutler, "DNSMOS P.835: A non-intrusive perceptual objective speech quality metric to evaluate noise suppressors," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Singapore (May 23–27, 2022), pp. 886–890, doi: 10.1109/ICASSP43922.2022.9746108.